Assignment-based Subjective Questions

1. **From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable?**

   Many of the categorical variables have been included in my model for the prediction like: holiday, mnth_Jul, season_Spring, mnth_sep. which are the dummies of month and season columns. We can see that their p-value is very less and it's very convenient for the model making as well. The VIF lies below 5 making it more favorable for our model.

2. **Why is it important to use drop_first=True during dummy variable creation?**

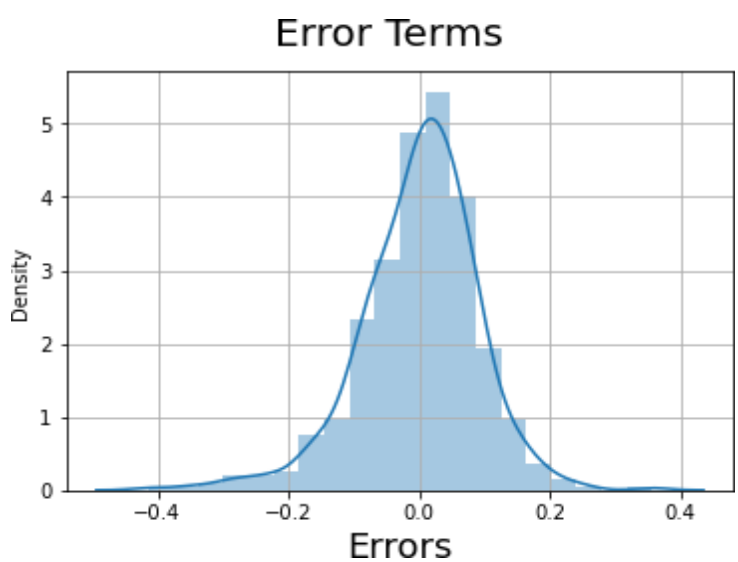
   drop_first=True is important to use, as it helps in reducing extra column created during dummy variable creation.
   Hence it reduces the correlations created among dummy variables.

3. **Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable?**

   "atemp" has the highest correlation with the target variable.

4. **How did you validate the assumptions of Linear Regression after building the model on the training set?**

   1. Linear relationship between X and Y
   2. Error terms are normally distributed (not X, Y)
   3. Error terms are independent of each other
   4. Error terms have constant variance (homoscedasticity)



Hence we can validate that “Error terms are normally distributed” after building the model on the training set.

5. **Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes?**

Based on the final model the top 3 features contributing significantly towards explaining the demand are:

A) Mnth_sep=0.570
B) Season_winter=0.47
C) temp = 0.45

So these are the variables play an important role in Bike sharing Maximum Booking.

General Subjective Questions

1. **Explain the linear regression algorithm in detail**.

Linear regression is one of the very basic forms of machine learning where we train a model to predict the behavior of your data based on some variables. In the case of linear regression as you can see the name suggests linear that means the two variables which are on the x-axis and y-axis should be linearly correlated.

Mathematically, we can write a linear regression equation as:

$y = mx+c$

Here, x and y are two variables on the regression line. m=
Slope of the line
c = y-intercept of the line
x = Independent variable from dataset y
= Dependent variable from dataset

2. **Explain the Anscombe's quartet in detail.**

Anscombe's quartet comprises four data sets that have nearly identical simple descriptive statistics, yet have very different distributions and appear very different when graphed. Each dataset consists of eleven points.

3. **What is Pearson's R?**

In statistics, the Pearson correlation coefficient (PCC), also referred to as Pearson's r, the Pearson product-moment correlation coefficient (PPMCC), or the bivariate correlation, is a measure of linear correlation between two sets of data. It is the covariance of two variables, divided by the product of their standard deviations; thus it is essentially a normalized measurement of the covariance, such that the result always has a value between −1 and 1.

The Pearson's correlation coefficient varies between -1 and +1 where:

- r = 1 means the data is perfectly linear with a positive slope (i.e., both variables tend to change in the same direction)
- r = -1 means the data is perfectly linear with a negative slope (i.e., both variables tend to change in different directions)

4. **What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling?**

It is a step of data Pre-Processing which is applied to independent variables to normalize the data within a particular range. It also helps in speeding up the calculations in an algorithm.

Most of the times, collected data set contains features highly varying in magnitudes, units and range. If scaling is not done then algorithm only takes magnitude in account and not units hence incorrect modelling. To solve this issue, we have to do scaling to bring all the variables to the same level of magnitude.

It is important to note that scaling just affects the coefficients and none of the other parameters like t-statistic, F-statistic, p-values, R-squared, etc.

Normalized scaling	Standardized scaling
Minimum and maximum value of features are used for scaling	Mean and standard deviation is used for scaling
It is used when features are of different scales	It is used when we want to ensure zero mean and unit standard deviation
Scales values between [0, 1] or [-1, 1].	It is not bounded to a certain range.
Scikit-Learn provides a transformer called MinMaxScaler for Normalization.	Scikit-Learn provides a transformer called StandardScaler for standardization.
It is an often called as Scaling Normalization	It is an often called as Z-Score Normalization
It is useful when we don't know about the distribution	It is useful when the feature distribution is Normal or Gaussian.

5. You might have observed that sometimes the value of VIF is infinite. Why does this happen?

If there is perfect correlation, then VIF = infinity. This shows a perfect correlation between two independent variables. In the case of perfect correlation, we get R2 =1, which lead to 1/ (1-R2) infinity. To solve this problem we need to drop one of the variables from the dataset which is causing this perfect multicollinearity

An infinite VIF value indicates that the corresponding variable may be expressed exactly by a linear combination of other variables (which show an infinite VIF as well)

6. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression?

Q-Q Plots (Quantile-Quantile plots) are plots of two quantiles against each other. A quantile is a fraction where certain values fall below that quantile.
For example, the median is a quantile where 50% of the data fall below that point and 50% lie above it.
The purpose of Q-Q plots is to find out if two sets of data come from the same distribution. A 45 degree angle is plotted on the Q-Q plot; if the two data sets come from a common distribution, the points will fall on that reference line.
If the two distributions being compared are similar, the points in the Q–Q plot will approximately lie on the line y = x. If the distributions are linearly related, the points in the Q–Q plot will approximately lie on a line, but not necessarily on the line y = x.

plots can also be used as a graphical means of estimating parameters in a location-scale family of distributions.

> A Q–Q plot is used to compare the shapes of distributions, providing a graphical view of how properties such as location, scale, and skewness are similar or different in the two distributions.